

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding requires identifying which event occurrence satisfies a query. When opposite event orders coexist, a model can localize a real sequence yet answer the wrong relation. We introduce RelTwin, a fine-tuning framework that contrasts locally valid timestamp answers. Shared excerpts form opposite-order windows in one recording; inverse queries exchange their target assignments. Candidate supervision and query-exchange consistency train relation-specific selection, followed by set-level preferences over coverage, count, and boundaries. Across SpotSound-Bench, Clotho-Moment, and UnAV-100, RelTwin improves both SpotSound-A and TimeAudio, with mIoU gains of up to 4.23 and 14.48 points over their published baselines. On controlled inverse-query development pairs, candidate supervision raises JointPairAcc by 31.25 percentage points over matched paired SFT. This advantage persists under altered silence timing, supporting improved query-window binding while inference retains direct timestamp generation.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query-window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing both categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query-window binding error*. Sequence supervision rewards each correct timestamp answer without explicitly contrasting it with another acoustically supported window under the same query.

RelTwin (Fig. 1) makes this competition explicit through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Both windows have real acoustic support, while their correctness as answers depends on the query. Reversing the query exchanges their

target assignments without changing the recording. RelTwin scores both timestamp answers under each query and trains the preference for the requested occurrence.

RelTwin uses the grounding model’s own timestamp-answer scores. Candidate supervision establishes the correct preference, and query-exchange consistency couples predictions after exchanging candidate labels. Set-level preference optimization then extends the comparison to coverage, event count, and boundaries. Inference remains direct timestamp generation.

We make three contributions. First, we characterize query-window binding errors when inverse relations coexist. Second, we formulate relation-specific grounding as competition between locally valid timestamp answers whose target assignments exchange with the query. Third, evaluations across backbones and benchmarks establish the adaptation benefit, while matched comparisons and timing intervention examine the query-binding mechanism.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions [3]. Audio-language models such as Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned text generation. TimeAudio introduces temporal markers and absolute time-aware encoding [7], while SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8]. RelTwin adapts grounding models to select among acoustically supported windows according to the queried relation.

Compositional audio-language learning. CLAP aligns audio and text in a shared representation space [9]. T-CLAP introduces order-reversed descriptions and a temporally focused contrastive objective [10]. CompA evaluates event order and attribute binding, and trains CompA-CLAP with composition-aware hard negatives [11]. CoSTALA combines spatio-temporal contrast with local event-caption alignment and feature consistency [12]. These methods learn compositional audio-text representations. RelTwin supervises relative scores of complete timestamp answers.

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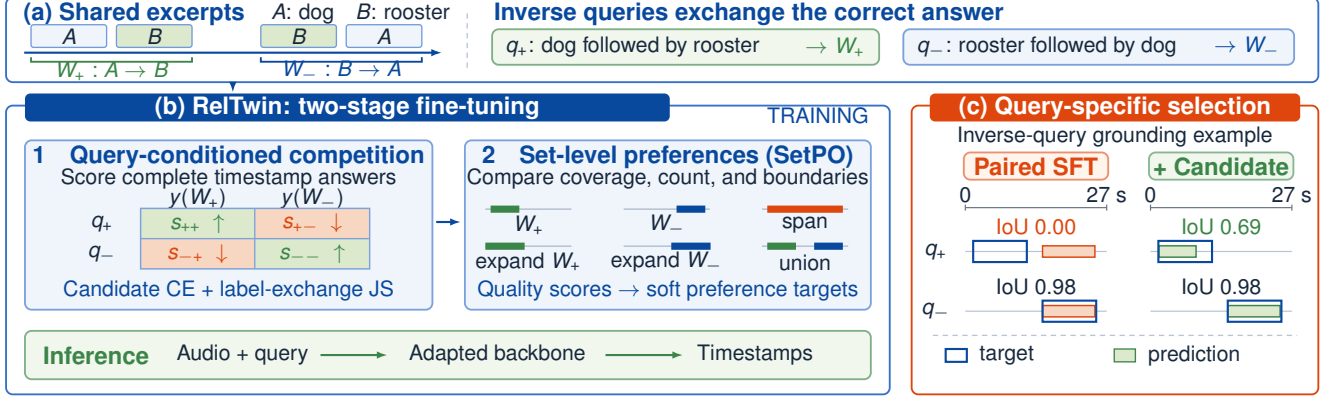


Fig. 1. RelTwin overview. (a) Shared excerpts hold acoustic content fixed while inverse queries exchange the target window. (b) Candidate supervision and query-exchange consistency train relation selection; SetPO extends the comparison to interval sets. Inference directly generates timestamps. Green/orange matrix cells indicate matching/competing answers. (c) A SpotSound-A inverse-query example compares paired SFT with candidate supervision. Blue outlines denote targets; filled bars denote predictions, with set-IoU above.

Hard negatives for temporal grounding. In compositional video grounding, SHINE combines generated hard-negative queries with within-video and cross-query saliency ranking [13]. RelTwin’s competing answer has real acoustic support but mismatches the requested relation. It becomes the correct answer when the query reverses, providing a paired contrast within a fixed audio context.

3. QUERY-SPECIFIC WINDOW LEARNING

3.1. Two true relations, two different answers

In one recording x , RelTwin arranges the same ESC-50 excerpts [14], A and B , as $A \rightarrow B$ and $B \rightarrow A$. Their local targets W_+ and W_- run from the first excerpt’s onset to the second’s offset, including the gap. Queries q_+ and q_- request the respective orders. Reusing the excerpts holds acoustic content fixed across windows, so their labels depend on the queried order: W_- is positive for q_- and negative for q_+ .

We alternate which target order appears first in the recording. Category rehearsal targets the union of both occurrences of A or B , retaining category-grounding supervision alongside relation selection.

3.2. Contrast timestamp answers during training

RelTwin scores complete timestamp answers under each query and trains their relative preference. We serialize an interval set S in the backbone’s native format: comma-separated spans for SpotSound-A and temporal tokens for TimeAudio. Answer tokens $y(S)$ exclude the prompt and receive the length-normalized score

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix $M_{ij} = s_\theta(x, q_i, W_j)$ for $i, j \in \{+, -\}$ indexes queries by rows and answers by columns; diagonal entries correspond to correct pairs (Fig. 1b). Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$. Candidate supervision explicitly trains the preferences $M_{++} > M_{+-}$ and $M_{--} > M_{-+}$. With row-wise probabilities $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$, $\tau > 0$, its loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

Query reversal exchanges the target assignment while preserving the recording. With label permutation Π , query-exchange consistency $\mathcal{L}_{\text{JS}} = \text{JS}(p_+, \Pi p_-)$ couples the paired preferences after relabeling. For rehearsal example (x_r, q_r, S_r) , define $\mathcal{R}_1 = -s_\theta(x_r, q_r, S_r)$. Stage 1 optimizes

$$\mathcal{L}_1 = \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{cand}} + \gamma \mathcal{L}_{\text{JS}}) + \beta \mathcal{R}_1. \quad (3)$$

Set-level preference optimization (SetPO). Selecting the relevant window also requires controlling its boundaries, coverage, and event count. Stage 2 extends answer competition to six interval sets: $W_\pm$, their expanded versions, their enclosing span, and their union. Quality axes are set-IoU, precision-recall margins above IoU 0.3/0.5, count agreement, and duration-normalized boundary accuracy. Utility v_{ij} averages mean quality and net Pareto dominance (number dominated minus dominating, normalized by candidate count minus one). This yields soft targets $t_i = \text{softmax}(v_i/\tau_q)$ and $\mathcal{L}_{\text{set}} = -\frac{1}{2} \sum_{i,j} t_i(j) \log p_i(j)$ over six answers, replacing $\mathcal{L}_{\text{cand}}$ in Eq. 3. Exchange Π swaps targets and their expansions, fixing the span and union.

Replay candidates perturb boundaries, add/drop events, merge spans, split the longest interval, or cover the recording. Stage 2 uses $\mathcal{R}_2 = \mathcal{R}_1 + \lambda \mathcal{L}_{\text{set},r} + \kappa \text{KL}(p_0 \| p_r)$, with frozen base-model distribution p_0 ; $\mathcal{L}_{\text{set},r}$ is the same listwise loss for one replay query. Inference directly generates timestamps.

Table 1. Temporal grounding across three benchmarks with RelTwin adaptation of TimeAudio and SpotSound-A (%; higher is better). Baseline scores are from SpotSound [8, Table 3]; Δ denotes the difference from the corresponding published backbone score. Bold marks maxima among listed methods.

Method	SpotSound-Bench			Clotho-Moment			UnAV-100 subset		
	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU
<i>Non-LLM models</i>									
WTATG	53.0	34.5	38.4	12.1	6.3	9.1	53.0	37.0	38.4
AM-DETR	25.5	15.0	19.5	89.8	88.0	80.9	59.0	46.0	42.8
<i>Proprietary models</i>									
Gemini-2.5-Flash	34.8	27.8	25.7	45.7	33.7	36.9	48.0	37.0	35.6
Gemini-2.5-Pro	20.0	17.3	19.7	40.4	32.5	32.5	39.0	35.0	34.6
<i>Open-source models and RelTwin adaptation</i>									
Kimi-Audio	3.0	1.3	3.2	0.7	0.1	0.9	8.0	2.0	5.3
Qwen2-Audio	5.5	1.8	4.4	6.1	1.3	5.7	14.0	4.0	9.7
Audio Flamingo 3	8.8	2.8	7.6	32.9	21.8	22.6	35.0	24.0	25.0
TimeAudio	7.8	1.8	10.2	39.5	24.9	28.6	21.0	7.0	16.0
TimeAudio + RelTwin	24.25	13.25	18.36	45.63	30.92	34.08	44.00	25.00	30.48
Δ vs. TimeAudio	+16.45	+11.45	+8.16	+6.13	+6.02	+5.48	+23.00	+18.00	+14.48
SpotSound-A	76.5	59.5	57.9	93.4	91.0	85.6	86.0	74.0	69.8
SpotSound-A + RelTwin	78.00	62.00	59.43	93.61	91.01	86.68	89.00	79.00	74.03
Δ vs. SpotSound-A	+1.50	+2.50	+1.53	+0.21	+0.01	+1.08	+3.00	+5.00	+4.23

4. EXPERIMENTAL SETUP

Data and evaluation protocol. Twin construction uses 40 ESC-50 classes and 256 recordings: 1,024 relation queries (512 pairs) and 512 category queries. The templates “followed by” and “and then” yield two inverse-query pairs per recording. Ten held-out classes provide 80 recordings, 320 queries, and 160 inverse pairs for relation development, with disjoint classes and original Freesound IDs at adaptation. Evaluation covers 400 SpotSound-Bench queries over 387 audios, 6,649 Clotho-Moment queries, and UnAV-100’s 100 queries over 77 audios [2]. We report set-mIoU and R1 at IoU 0.3/0.5, clipping intervals to duration.

Joint localization and relation selection. For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G| / |S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, let $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. Inspired by paired compositional evaluation [15], it distinguishes query-specific selection from coverage: returning the union of two disjoint, equal-length targets passes PairAcc (IoU 0.5) but fails JointPairAcc. We report pair-level percentages; same-answer counts require identical parsed interval lists.

RelTwin training. TimeAudio adaptation starts from its official checkpoint, with 384 Stage 1 updates followed by 128 SetPO updates. Its 31.76M trainable parameters use learning rates 2×10^{-6} (LoRA [16]) and 5×10^{-6} (temporal/audio connectors). Event/relation twins mix 3:1; replay mixes Clotho training, synthetic scale/cardinality, TEMPO

training, and AudioGrounding training at 4:3:2:1. We use $\lambda = \gamma = 1$, $\beta = 0.75$, $\kappa = 0.2$, $\tau = 1$, $\tau_q = 0.15$, and 15% interval expansion (minimum 0.05 s).

Matched mechanism experiment. Table 2 isolates $\mathcal{L}_{\text{cand}}$ on SpotSound-A: Eq. 3 with $\gamma = 0$, $\beta = 0.5$, and $\lambda = 0$ (paired SFT) or 1 (candidate supervision), evaluated after Stage 1. Both conditions share initialization, relation/rehearsal data, data order, and 256 AdamW [17] updates at 5×10^{-6} , with zero weight decay and gradient clipping 1. LoRA has rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Each update accumulates one pair and one rehearsal query. Candidate supervision adds four gradient-free scores and weighted gradient recomputation to SFT’s two positive-answer backpropagations.

Inference. We use greedy one-beam decoding, capped at 128 new tokens. The mechanism experiment uses 16 kHz mono audio and bfloat16; timing tests use its frozen checkpoints (Section 5.3).

Comparison and statistics. Table 1 uses published baseline scores; Table 2 evaluates the backbone and training variants under shared inference. Relation-development and public benchmark scores informed development. TimeAudio uses seed 0. Mechanism point estimates use seed 0, selected by SpotSound mIoU among five runs, with matched SFT; Table 2(c) separately reports seeds 0/1/2 as mean $\pm$ sample SD.

We use 20,000 audio-group bootstrap resamples for 95% CIs; relation tests resample 39 connected source-sharing groups (largest: 18 audios). The intervals quantify sampling variation for the selected checkpoints.

Table 2. Mechanism experiment: isolating RelTwin’s candidate supervision on SpotSound-A. (b) Matched data and updates; (c) mean $\pm$ sample SD. Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
Candidate supervision	59.43	88.55	84.38	7.50	1/160
Gain over paired SFT	+1.02	+13.97	+31.25	−32.50	−36 pairs
<i>(c) Candidate supervision: three training seeds</i>					
Candidate supervision	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

5. RESULTS AND ANALYSIS

5.1. Grounding performance across benchmarks

RelTwin improves mIoU for both backbones on all three benchmarks (Table 1). SpotSound-A gains 1.53, 1.08, and 4.23 points on SpotSound-Bench, Clotho-Moment, and UnAV-100, respectively; TimeAudio gains 8.16, 5.48, and 14.48 points. The improvements extend across both model families and datasets, with the highest absolute scores attained by adapted SpotSound-A among the listed methods.

On 400 SpotSound queries, candidate supervision improves mIoU by 1.02 points over matched paired SFT (95% audio-group bootstrap CI: [0.23, 1.81]). Its three-seed mean is 59.16 (Table 2c).

5.2. Timestamp contrasts recover query-specific selection

Candidate supervision recovers relation-specific answers, raising JointPairAcc from 53.13 to 84.38 (Table 2). Identical-answer pairs fall from 37 to 1 out of 160; the original 37 span 22 recordings. Of these collapsed pairs, 27 become jointly correct, with both queries selecting their requested relations. Figure 1 illustrates this recovery.

The JointPairAcc gain is 31.25 points (source-group CI [22.35, 40.50]), accompanied by a 32.50-point reduction in Swap error. Across all pairs, 51 become jointly correct and one regresses. Relation mIoU increases by 13.97 points (CI [9.84, 17.51]), showing that improved relation selection also increases temporal overlap.

Across three seeds, mean JointPairAcc is 81.46; same-answer counts are 1, 0, and 0.

5.3. Binding gains persist after timing intervention

We test whether the binding advantage persists when silence durations change while event waveforms, order, and queries remain fixed. We recompose 80 development recordings with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between

windows, and 2 s tail. The frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

On the 40 modified recordings, candidate supervision retains a 20-point JointPairAcc advantage over paired SFT (87.50 vs. 67.50; CI [2.94, 33.33]), compared with original scores of 85.00 and 42.50. The advantage persists under the changed silence layout. Across all 160 queries, SFT and candidate supervision score 81.34 and 90.65 mIoU.

5.4. Why locally valid answers matter

Both candidate windows contain the queried categories, yet inverse queries require different answers. Reduced same-answer collapse and higher JointPairAcc distinguish improved relation selection from event recognition in the controlled test. Candidate supervision trains this distinction by placing acoustically supported windows in direct competition under each query.

6. CONCLUSION

RelTwin learns query-specific grounding by contrasting timestamp answers with real acoustic support. Inverse queries exchange the target assignment within a fixed recording, making relation-specific selection an explicit training objective. Both adapted backbones improve mIoU over their published baselines on three benchmarks. Matched candidate-supervision experiments and timing intervention support improved query–window binding, while inference retains direct timestamp generation.

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Compliance with Ethical Standards. We use public ESC-50 [14], SpotSound [8], Clotho-Moment, UnAV [2], TEMPO, and AudioGrounding data, including programmatically constructed sequences. The study involved no new recordings, participant recruitment, human or animal experiments, or listening studies.

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